HTE operator handbook

Local OT-2 dual-ligand fluorination screen

Edition: 11 October 2026 (Europe/Zurich). Generated from editable repository sources.

HTE configuration review - bench checks remain open

Campaign	66 first-round wells; 10 second-round wells
Reaction	5 µmol substrate; 50 µL toluene; 95 °C; 240 min per round
Catalysts	6 mol% each ligand; 12 mol% Pd(COD)(DQ)
Sample dilution	20 µL extract + 180 µL IS-free LC diluent
Software evidence	101 local tests; 0 failures. Rehearsal chemistry and Claude responses are synthetic.
Bench readiness	9 applicable validation fields still unconfirmed; exact hardware and analytical settings must be supplied by HTE.
Provisional timing	729 min planned / 720 min competition limit

How to keep this handbook current

Edit the Markdown sources in **docs/** and the relevant campaign configuration, then regenerate locally or download the fresh GitHub workflow edition. The PDF is a dated export; the editable sources remain the master. Keep the PDF with its build manifest to check for later source changes.

```
python scripts/build-hte-pdf.py
```

Open the shared repository - see PDF update instructions.

Generated quantities from the real selected campaign take precedence over the synthetic demo. This handbook includes the procedures and outstanding decisions; it does not supply unmeasured geometry or analytical settings.

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1. Quickstart and responsibilities

Editable source: docs/start-here.md

Send the organizers the HTE configuration checklist and give the experimental operators the **two-page staff checklist**. The complete runbook remains a technical reference. The checklist requests the unresolved facts; the runbook explains stock preparation, deck loading, each reaction/analysis stage and the automated decision.

What to ask the organizers

1. **Reaction hardware:** Which sealed glass-vial array, insert and seal can they supply and securely mount on the OT-2 heater-shaker? Ask for the measured labware JSON, validated 95 °C heating/shaking/sealing, actual thermal lag and a second vial array for predosing round two. Para-Dox 104960 is a candidate, with compatibility pending.
2. **LC method and data:** Ask for the exact method, measured full two-minute injection cycle, quantitative channel/wavelength, retention times, injection volume and a representative CSV/XLSX export. Include well/sample mapping, failed injections, nondetections and an end-of-sequence signal. Raw DAD data are needed only for optional MOCCA processing.
3. **Extraction and calibration:** Confirm recovery and homogeneous sampling from the toluene/water/ACN mixture, acceptable dilution/concentration, calibration and naphthalene suitability, and the LC plate/seal. Validate naphthalene added with SM through drying/heating before treating its area as a yield reference. Schedule five nonzero analyte levels at fixed IS, blanks and independent checks while the first reaction runs; use the stock and calibration instructions.
4. **Stocks and liquid handling:** Confirm identity/assay, solubility of every proposed stock, glovebox stock preparation, containers/dead volumes, solvent pipetting and acceptable source-dedicated tip reuse. Validate aspiration heights and the dry endpoint for ambient second-round DCM evaporation.
5. **Time and access:** Request a timed rehearsal, continuous LC availability, local OT-2 App access, a shaker for the analytical plate and a workstation allowed to contact Anthropic. Stock preparation and initial selection are outside the clock; the final ten analyses end it.

The chemistry and two-minute analyses already occupy 632 of 720 minutes. The complete provisional plan is 729 minutes; the organizers need to help establish a measured plan that fits.

What HTE follows

Stage	Operator action	Software handoff
Before the clock	Confirm equipment/method, prepare stocks in the glovebox, calibrate handling and rehearse	Campaign configuration; real first-batch selection; printed stock and deck-load CSVs
First assembly	Dose combined SM/IS into 66 wells, remove DCM, add ligands/Pd/toluene, seal and begin the four-hour hold	Local first-round dosing/reaction protocol
During first hold	Measure authentic standards/calibration; predose ten second-round SM/IS wells and allow ambient DCM evaporation	Calibration CSV and ten-well predose manifest
First workup and LC	Cool, work up, dilute and measure all 66 samples; publish the complete export	Local workup protocol; peak CSV plus completion marker
Decision	Review QC and the ten new pair assignments	Watcher produces `next_design.csv` and `next_dosing.csv`
Second assembly	Verify predosed wells are dry; add the selected catalysts and toluene, seal and hold four hours	Local second-round protocol; substrate and IS are not dosed again
Final workup and LC	Work up and measure all ten samples; stop the clock	Final calibrated results and logs

Staff follow the short checklist; the runbook supplies the detailed reference. Exact glass hardware, extraction and analytical settings are still pending; they must be resolved before physical execution.

Where to put the Claude API key

Use the key on the **operator laptop**. Create `anthropic.key` in the repository root by copying `anthropic.key.example`, then replace its single placeholder line with your Anthropic API key. Paste only the key, with no quotes or `ANTHROPIC_API_KEY=` prefix. Scoring and the watcher read this file automatically when run from the repository root. The robot does not need it.

`anthropic.key` is excluded from Git and the HTE handoff. Keep it on the operator laptop. An existing `ANTHROPIC_API_KEY` environment variable takes precedence; `.env` files are not automatically loaded.

From the repository root, create/activate a Python environment and install the package:

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -e .
cp anthropic.key.example anthropic.key
# Open anthropic.key in a text editor and replace the placeholder.
zsh scripts/first-batch.zsh
```

The script uses the local key file and runs five real scoring calls over the 465 pairs, followed by the 50/50 score/diversity selection. Its output is `output/first-batch-<timestamp>/first_batch.csv`: **65 pair assignments plus one L17 tBuBrettPhos reference**, including well and sample IDs. All 31 ligands must occur in the pair wells; the single reference does not count toward coverage. `first_batch.csv.ligand_coverage.csv` lists each ligand and its pair-well count, and the selection trace flags any coverage-driven Claude-cutoff override. Scoring files preserve the exact prompts, individual scores, token usage and the cost preflight. Responses contain scores only; extended thinking is disabled.

These are paid calls within the configured estimated $20 campaign cap. No key belongs in a committed file or chat message.

The script accepts an output folder and campaign path, for example `zsh scripts/first-batch.zsh output/real-first-batch local_data/campaign.json`; an optional third argument supplies the inventory path. Use that same campaign file throughout export and LC feedback. The first batch can be selected before the organizers finalize hardware and LC details. Protocol/stock exports use the confirmed campaign configuration as described in the runbook. Do not use the synthetic demo ranking for experiments.

Review the structured prompt and SI evidence before scoring. To inspect the exact current API requests without a key or paid calls, run `python -m hte.cli preview-prompts --output output/prompt-review`. Scoring also saves the exact request JSON and a readable prompt beside every response.

The later watcher can read the same `anthropic.key` file. As an alternative, set the key for a terminal with hidden input:

```
read -rs 'ANTHROPIC_API_KEY?Claude API key: '
export ANTHROPIC_API_KEY
```

If no key file or environment variable exists, the first-batch script asks for hidden input. A prompted key stays in that process; it is not saved to disk or set in the parent terminal. Real model access is checked before scoring, and access failures stop the campaign without silently selecting another model.

Prompt caching is already enabled for both rounds. Shared context uses a one-hour lifetime, and the score command prints the provider-reported cache status. Inspect `scoring/cache_summary.json` for read/write tokens and estimated cost. No additional key setting is needed.

Fake first feedback loop

Install the test dependencies if they are not present, then run:

```
python -m pip install -e '.[test]'
python -m hte.cli rehearse-loop --output output/fake-first-loop
```

Use a fresh output directory for each rehearsal. This uses the real frozen T5 pair embeddings, **synthetic LC results**, and an explicitly mocked Claude client. It makes zero paid calls and needs no key. It runs both scoring campaigns through the real scoring code and the first-round event through the real watcher.

The report verifies 66 first-round measurements reach the scorer, only 400 untested pairs are rescored, ten new pair assignments are generated, substrate is not dosed again, incomplete exports are ignored and repeated completion events do not trigger more calls. `rehearsal_report.json` points to the resulting `next_dosing.csv`. Scores and yields from this exercise demonstrate data flow only.

What has run from GoLLuM

Real T5-base inference generated the included 465 × 768 pair embeddings, averaging both ligand orders. Initial selection uses those embeddings for diversity, following the language-model representation idea that motivated GoLLuM. With no measured pair yields, it does not fit a yield model.

The attributed upstream GoLLuM optimizer has been executed with a local GP-trained projection over frozen embeddings and synthetic observations. This adapter keeps the original language-model weights frozen; it is not the full upstream fine-tuning pipeline. It is optional and is not used to decide the approved all-LLM second round. For this campaign, GoLLuM is used only as inspiration for embedding-based initialization. After the first yields arrive, Claude alone scores/selects the second round; no GoLLuM yield model is trained or consulted.

2. Complete operator runbook

Editable source: docs/hte-runbook.md

This package screens 65 distinct ligand pairs plus one tBuBrettPhos reference, then 10 new pairs. It converts completed LC peak exports into calibrated yields, sends those results to five repeated Claude scoring calls, and generates the next dosing CSV. Robot operation stays local through the Opentrons OT-2 App. A workstation with permitted internet access makes the Anthropic calls; the robot never needs internet or remote control.

Software verification is separate from bench readiness. The provided demo is synthetic and its protocols refuse physical execution. The real campaign still needs an exact glass-block definition and mounting/seal validation, solvent delivery checks, a homogeneous extraction, and the HTEL analytical method. The real first-round selection must be generated with the operator's Claude key; no synthetic ranking should be used for chemistry.

Fixed experiment and interpretation

Item	Approved campaign setting
Substrate	Pyridine-2-sulfonyl fluoride / PyFluor, `FS(C1=NC=CC=C1)(=O)=O`
Product standard	2-Fluoropyridine, `FC1=NC=CC=C1`
Amount / concentration	5 µmol / 0.1 M in 50 µL final toluene solution
Pair ligands	0.30 µmol each, 6 mol% each; loading counts ligand molecules
Pd	0.60 µmol Pd(COD)(DQ), 12 mol%; one Pd per complex
Reference	L17 tBuBrettPhos, 0.60 µmol / 12 mol% total ligand; same Pd and conditions
Temperature / hold	95 °C / 240 minutes after the module reaches the setpoint
Atmosphere	Reaction assembly under air; ligand stocks prepared/stored in glovebox
Additional reagents	0.050 µmol naphthalene analytical IS added with SM; no base or external fluoride source
Round 1	65 pairs + 1 reference = 66 wells
Round 2	10 untested pairs; no reaction repeats or additional controls
Objective	Calibrated product yield; conversion is diagnostic

The reference is a fair, resource-efficient comparison against a good single ligand at the same temperature, total ligand loading and Pd loading. An improvement over this reference supports a better mixture under these conditions. It does not prove synergy relative to both individual partners, reproducibility, or a broader substrate scope. Those claims require subsequent partner controls, repeats and scope experiments. Score SD measures disagreement among model calls, not experimental error.

The supplied Chemical Science SI Table S2, p. S35, contains 18 single-ligand yields for this substrate. These are captured in `hte_inputs/literature.json`, including tBuBrettPhos 77%, AdBrettPhos 78%, MorDalPhos 77%, Me4tBuXPhos 72%, RockPhos 63% and Me3OMe-tBuXPhos 61%. They are high-temperature qualitative priors, not 95 °C measurements. The SI's air example uses a different Pd

source at 150 °C. Do not merge these data with the main-text ligand table or train a 95 °C pair surrogate on them. Paper and SI.

The thesis benchmark motivates repeated mechanism-informed scoring, but its dual-ligand benchmark is decarbonylative Suzuki coupling. Ligand relay in the fluorination remains a hypothesis. The Nature multicatalysis paper is contextual inspiration rather than this campaign's measured training data. Nature paper.

Before the competition: resolve hardware and analytics

Use the HTE configuration checklist to obtain the remaining facts. The start-here guide explains responsibilities and hidden API-key entry; rehearsal results document the fake feedback loop. Record confirmed facts in a campaign-specific copy of `hte_inputs/campaign.json`. Keep that copy, inventory, calibration, protocol exports and run logs together. The production commands below use `local_data/campaign.json`; create it once and edit it as HTE confirms the settings:

```
mkdir -p local_data
cp hte_inputs/campaign.json local_data/campaign.json
```

Do not recopy the template over a configured campaign. Initial selection can run with provisional hardware fields; live protocol export requires the released configuration.

HTEL publicly lists a Thermo Vanquish Horizon Duo with ISQ-EM, tandem column operation, and a DAD detector. UZH's Chromeleon page lists `LC-ISQ-HTL-01`. The exact two-minute method, quantification wavelength, retention times, injection volume and export headers are not public. Prefer a validated DAD calibration for yield and use MS to confirm peak identity. Naphthalene may be unsuitable for HESI quantification: do not assume an MS internal-standard signal. HTEL equipment, Chromeleon instruments.

If a measured thermal lag requires additional equilibration after the module reaches 95 °C, set `chemistry.thermal_equilibration_minutes` to that validated time. The generator adds it before starting the 240-minute hold. Include both rounds' extra equilibration in the measured heating-overhead estimate. The initial zero is an unresolved bench placeholder, not proof of instant equilibration.

A plausible glass option is the Para-Dox Gen II 96-position parallel-synthesis block 104960 with 8 × 30 mm, 1 mL glass inserts, preferably the manufacturer's assembled tray 884001. Its drawing gives 127.8 × 85.5 mm, 9 mm pitch, and approximately 46.2 mm assembled height. The footprint matches ordinary microplates; this alone does not establish heater-shaker compatibility. Block drawing, regular microplate dimensions.

HTE must check the complete block on the supplied universal flat adapter: latch engagement, closed lid/bolt clearance, mass during shaking, stable contact, vial seating, thermal performance and sealing with toluene at 95 °C. A module setpoint does not prove a large aluminum block's liquids are at that temperature. Validate the thermal lag with a representative filled vial and include the resulting procedure in the hold timing. The 1 mL vial allows the 100 µL post-quench volume, but 50 µL is shallow; aspiration immersion needs particular attention.

Use a combined block-plus-vial custom Opentrons JSON with measured **open-block** height, well opening diameter, depth, vial-bottom Z and A1 position. Do not substitute a PCR definition for glass. The manufacturer's drawing does not establish all internal vial dimensions or measured deck offsets. Set `reaction_definition_path` to that JSON, `reaction_labware` to null, and the true working volume. If a custom adapter is needed, the supplied adapter loading code must be reviewed for that adapter before use. A block that cannot be latched securely cannot be shaken by this protocol. For custom labware, contact johannes.schoergenhumer@chem.uzh.ch.

The well coordinates describe the open vials, but travel clearance must also account for the sealed block's maximum height. HTE must check both states; the model cannot treat a removed lid as proof of clearance while a sealed block is on the heater.

Follow the manufacturer's PFA-film/rubber-mat stack and cross-pattern sealing instructions, including the specified torque for the exact reactor. Validate solvent loss for a four-hour hold and gas/pressure compatibility: this reaction releases SO2. Recheck after cooling before opening. Ordinary plate seals are not evidence of a suitable reaction seal. Assembly instructions.

Two reaction vial arrays are required because round two is predosed while round one reacts. Only one heater is required. HTE should confirm whether a second assembled block or a compatible removable vial tray can be used for ambient evaporation and later returned to the validated reaction block without changing well mapping.

For the LC plate, use the appropriate library definition for HTE's PP plate, substituting the exact brand/geometry as necessary. `nest_96_wellplate_2ml_deep` is a representative placeholder, not an instruction to use a 2 mL plate. A validated 200 µL working-volume PP plate may be more economical, provided its actual working volume, mixing, autosampler access, seal and needle-height requirements are satisfied. Avoid PS for this solvent-containing workflow unless HTE has verified compatibility.

Install and rehearse locally

Use Python 3.12 on the operator workstation, not the robot's internal Python. From the package root:

```
python3.12 -m venv .hte-venv
source .hte-venv/bin/activate
python -m pip install -e '.[sources,test,analytics,embeddings]'
# Optional embedding rebuilds and historical adapter tests:
python -m pip install -r requirements-gollum.txt
python -m pytest -q tests/hte
python -m hte.cli demo --output output/hte-rehearsal
```

`requirements-hte-tested.txt` pins the principal versions used here; for the same principal package versions, install it before `python -m pip install -e .`. It is a tested version list, not a complete lock of all transitive dependencies.

The demo supplies a synthetic LC method and a library PCR plate only for software simulation. It has 66 first-round and 10 second-round samples, valid nondetection bounds, no repeated pairs, and no second substrate dose. Never use its selections, calibration or labware as real experimental inputs.

Simulate each of its four Python files with the pinned OT-2-capable package:

```
python -m opentrons.simulate output/hte-rehearsal/round1/dose_and_react.py
python -m opentrons.simulate output/hte-rehearsal/round1/workup.py
python -m opentrons.simulate output/hte-rehearsal/round2/dose_and_react.py
python -m opentrons.simulate output/hte-rehearsal/round2/workup.py
```

Opentrons 8.8.2 is pinned for workstation tests because the installed 9.1.2 simulator rejected OT-2 protocols. Protocol files use API 2.18. HTE must also analyze them in its actual local OT-2 App/robot software, confirm the version supports this API, and perform Labware Position Checks. Simulation checks command compatibility and deck restrictions, not glass fit, organic-solvent volume accuracy, dryness, phase behavior or temperature in the liquid.

Do a measured rehearsal of both dosing/workup stages with the final hardware and liquids or appropriate validated surrogates. Record real heating/cooling, manual handling and injection-cycle times. Test Claude model access and a scoring campaign before the competition. Do not discover a quota or schema issue during the second-round decision.

Inventory, embeddings and first-round selection

`hte_inputs/ligands.csv` contains all 31 workbook ligands, source locations and molecular weights. The original workbook remains unchanged. Two user-authorized corrections are recorded in the CSV: L11 now contains the methoxy oxygen; L27 is tris(3-chlorophenyl)phosphine. Commercial Me3(OMe)tBuXPhos can be a regioisomer mixture; preserve the actual supplier/lot identity rather than claiming a pure regioisomer from one representative SMILES. Original ligand study.

There are 465 unordered distinct pairs. The included `hte_inputs/pair_embeddings_t5-base.npz` contains real, frozen T5-base text embeddings, 768 dimensions per pair. Both ligand orders are encoded and averaged. Provenance and model revision are saved beside it. This uses names, structures and the fixed reaction context, without manually calculated steric/electronic descriptors. It is inspired by the language-model representation approach in GoLLuM, and is not a trained GoLLuM yield model. Rebuild after changing ligand identities/names or inventory metadata:

```
python -m hte.cli embeddings --model google-t5/t5-base --download \
  --output output/hte-embeddings/rebuilt.npz
```

With no measured pair yields, GoLLuM contributes its language-model representation approach rather than a trained yield model. Our frozen T5-base encoder embeds the names, SMILES and reaction context for all 465 pairs; averaging both ligand orders makes each representation symmetric. Claude independently scores every pair five times. Software converts the mean scores to rank percentiles, then combines 50% Claude rank and 50% embedding-diversity rank at each greedy selection. Initial diversity uses distance from the embedding centroid; later diversity uses minimum distance from selected pairs. Equal scores/distances receive average ranks, so ligand IDs do not create a preference between ties. These ranks are selection preferences, not calibrated chemical probabilities. Initialization needs no measured pair yields: upstream initializers select from input representations. After round one, only Claude receives measured feedback and selects ten new pairs; GoLLuM is not used again.

All 31 ligands must appear in at least one of the 65 first-round pair wells. The L17 single-ligand reference does not satisfy this pair-coverage requirement. Coverage is secured first: eligible pairs that introduce two unseen ligands are preferred where possible, and the same 50/50 consensus chooses between them. Once all ligands are represented, ordinary consensus filling completes the 65 pairs. The usual Claude percentile floor is 0.25; a coverage choice may go below that floor when necessary, and its override is recorded. This requirement adds no wells, singles or repeats.

`*.selection.csv` records the two ranks, coverage phase, newly covered ligand IDs and any floor override. `*.ligand_coverage.csv` lists each ID/name and number of pair wells; verify 31 covered and no zero counts before stock preparation. The metadata also records coverage and overrides. Design creation, dosing export and LC feedback processing reject missing required ligands. Keep `design.require_all_ligands_in_round1_pairs = true` in the campaign configuration. Conditions are randomized across occupied wells in column-major order to use full eight-well columns efficiently.

Store the Claude key on the operator workstation. Copy `anthropic.key.example` to `anthropic.key` in the repository root and replace the single placeholder line with the raw key, without quotes or an `ANTHROPIC_API_KEY=` prefix. Scoring and the watcher automatically read this file when run from the repository root. The file is ignored by Git and excluded from this handoff. It must not appear in a protocol, CSV, notebook output or committed file.

An existing `ANTHROPIC_API_KEY` environment variable takes precedence over the file; `.env` files are not automatically loaded. On macOS zsh, hidden entry can set that variable without putting the key in command history:

```
read -rs 'ANTHROPIC_API_KEY?Claude API key: '
export ANTHROPIC_API_KEY
```

Run scoring or the watcher from the repository root. The local key file works on other platforms too; use HTE's approved secret setup if required by the workstation policy.

```
python -m hte.cli --config local_data/campaign.json score --output output/hte-round1-scoring
python -m hte.cli --config local_data/campaign.json \
  design --ranking output/hte-round1-scoring/ranking.csv \
  --embeddings hte_inputs/pair_embeddings_t5-base.npz \
  --output output/hte-round1-design.csv
```

Five calls independently score every pair after shuffled candidate ordering. The scoring prompt guide lists all 31 IDs/names/SMILES and 18 CAS-matched SI Table S2 yields. The prompt names pyridine-2-sulfonyl fluoride (PyFluor) and 2-fluoropyridine, asks about potential ligand cooperativity and scores expected relative product-yield performance. Only individual literature yields and their brief condition label are sent; the 13 unreported values remain unknown. An explicit catalog maps every pair ID to its two ligand IDs.

Run `python -m hte.cli preview-prompts --output output/prompt-review` to inspect the exact requests without API calls. Paid scoring saves the same readable prompts and request JSON beside each response. The parser accepts only `scores`, with one finite numeric 0-100 score for every requested ID. It rejects reasoning fields, missing/duplicate/extra IDs, numeric strings, nulls and booleans. Extended thinking is disabled. Raw responses, usage and repeated scores are retained. Fixed experimental conditions remain configured by the operator.

Prompt caching is enabled by default with a one-hour TTL in both rounds. Shared chemistry context and measured feedback are cached; shuffled candidate IDs remain outside the cached prefix. Stable schemas and a first scoring response before parallel followers preserve reuse. Each scoring directory contains `cache_summary.json` and response cost breakdowns; inspect the provider's read counters to verify live hits. Cache expiry across the reaction requires a fresh write in round two. The budget preflight includes the write premium and assumes no cache hits.

Claude Opus 4.8 is explicitly configured; the code checks access and never silently substitutes a model. At the verified $5/M input and $25/M output token prices, five requests each using the full 10,000-output-token allowance cost at most $1.25 for output, before input and retries. A $20 cap per campaign is therefore plausibly ample for the present prompts, but the code performs a token-count preflight, reserves one retry, and refuses a campaign whose estimated bound exceeds the cap. The two rounds have separate $20 caps; this is not a shared $20 project budget. A preflight is an estimate, not a provider-enforced spending limit; use a provider limit too if required. Model and pricing.

Resume the same scoring directory only with identical inputs; successful calls are reused. An interrupted request may have been charged even when no response was saved. A saved attempt or unvalidated raw response blocks automatic repayment: inspect it and deliberately start a new campaign if a replacement call is needed. OS locks prevent concurrent scoring/watchers on the same directory and release after a crash; the empty lock file can remain.

Stock preparation before the clock starts

Once exact hardware is configured, export for simulation/review without `--live`:

```
python -m hte.cli --config local_data/campaign.json export --design output/hte-round1-design.csv \
  --output output/hte-round1-review
```

After completing the bench checks, LC settings and measured timing, release the actual protocols into a separate folder:

```
python -m hte.cli --config local_data/campaign.json export \
  --design output/hte-round1-design.csv --output output/hte-round1 --live
```

Verify the first-batch ligand coverage CSV, then print `stock_preparation.csv`, `deck_loads.csv`, `dosing.csv`, `round2_predose.csv`, `lc_sequence.csv` and `tip_budget.json`. **Use that campaign's generated quantities**, not the synthetic demo's quantities. `stock_preparation.csv` specifies total solution to prepare; `deck_loads.csv` specifies how much to place at each source for each protocol. The remaining prepared stock stays capped off deck. Source capacities are checked; large solvent reserves are not all loaded into a single well.

The stock planner includes first-round consumption, tip surplus, a conservative reserve for any ligand to be selected throughout round two, 50% extra usable stock and dead volume. Default dead volumes are 100 µL per 1.5 mL tube and 1,000 µL per reservoir well. These are provisional aspiration allowances and need measurement with the actual source geometry. The second-round substrate is already included in first-round stock consumption. The ledger column `protocol_consumption_ul` reports consumption for the exported round, including any first-round predosing. Keep enough fresh tips/racks for the printed budget plus contingency; a source-dedicated reuse setting must be validated before physical execution.

Stock	Concentration / solvent	Per pair well	Preparation basis
PyFluor + naphthalene	250 mM SM + 2.5 mM IS in DCM	20 µL before evaporation	5 µmol SM + 0.050 µmol IS; prepare at least 5 mL
Each ligand	30 mM in toluene	10 µL each	0.03 × molecular weight in mg/mL
Pd(COD)(DQ)	60 mM in toluene	10 µL	22.728 mg/mL, using MW 378.80
Additional toluene	Neat	20 µL	Final reaction solvent totals 50 µL
Reference ligand L17	Same 30 mM stock	20 µL	Other ligand volume zero
Workup reagent	1:1 water/ACN, 1% v/v HCOOH, **without naphthalene**	50 µL	IS was added with substrate
LC diluent	1:1 water/ACN, 1% v/v HCOOH, **without naphthalene**	180 µL	Add before the sample aliquot

Bring solutions to their final volume after weighing. If the COA gives an applicable assay fraction, enter it in `stock_preparation.assay_fractions`, keyed by ligand ID, `Pd`, `substrate_DCM` or `naphthalene`; weighed mass is then corrected by dividing by that fraction. Without a supplied fraction the code assumes 1.0. Check molecular weights against the actual bottle identity, including salts/adducts/solvates; a SMILES-derived weight is not a substitute for a COA.

The `substrate_DCM` row reports PyFluor in `stock_mM`/`mass_mg` and the second solute in `naphthalene_stock_mM`/`naphthalene_mass_mg`. The `workup` row now reports zero naphthalene. Prepare the combined 5 mL stock in a capped 20 mL glass vial, using an accurately measured naphthalene master rather than directly weighing approximately 1.6 mg. Load the generated quantity into reservoir A1, approximately 2.54 mL with the default reuse plan, and retain the rest capped. The combined-stock and calibration instructions specify the recipe, containers, five analyte levels with fixed IS, and separate authentic/analytical stocks.

Open ligands and prepare stocks in the glovebox. Label ID, name, CAS, concentration, solvent, assay correction, lot, preparation date, and rack position. Use suitably sealed compatible containers and return capped reserves to inert storage promptly after dispensing, preserving them after the hackathon. Pd(COD)(DQ) identity is distinct from Pd2(dba)3; verify the bottle and its purity. Manufacturer identity.

All stocks must be homogeneous at dispensing temperature. A proposed concentration is not a solubility measurement. Especially verify 60 mM Pd and every 30 mM ligand in toluene. If any stock cannot be prepared, change its concentration and recompute volumes through a reviewed code/configuration change; the current implementation uses one common ligand concentration. Do not pipette a suspension, aspirate repeatedly to force material through a tip, or add an unrecorded cosolvent. Particles and bubbles can cause unknown delivered concentrations. Stop, inspect, establish the actual dissolved concentration and regenerate the affected plan before proceeding.

Deck and liquid locations

Left mount: p20 single GEN2. Right mount: p300 eight-channel GEN2. The same setup is used in both rounds.

OT-2 slot	Contents
1	Opentrons 24-position 1.5 mL tube rack; L01-L24 at exact CSV addresses
2	12-well reservoir; A1 combined SM/naphthalene/DCM, A2 Pd/toluene, A3 neat toluene, A4 workup solvent without IS, A5 LC diluent without IS
3	Open reaction block for dispensing/sampling; returns here for off-module cooling
4	Second 24-position tube rack; L25-L31 at exact CSV addresses
5	p20 tip rack
6	First dosing protocol: second-round substrate array; workup protocols: LC plate
7	Empty
8	p300 tip rack
9	p300 tip rack
10	Heater-shaker with validated standalone adapter
11	Empty

Load only the reagents used in that stage, according to `deck_loads.csv`. Source addresses do not change between stages. The tube rack is the library `opentrons_24_tuberack_eppendorf_1.5ml_safelock_snapcap`; reservoir is `nest_12_reservoir_15ml`; substitute actual brands only with verified definitions. Remove tube caps for dosing and recap afterward. Do not place uncovered solvent stocks next to the hot block for four hours; remove/cap sources once dosing is finished at the appropriate paused stage.

Slots 7 and 11 remain empty because adjacent eight-channel access is restricted. Slot 8 is a tip rack, which is the documented exception immediately in front of the heater-shaker. All block transfers are manual through `move_labware(..., use_gripper=False)` prompts. Do not add a gripper command on OT-2. Heater-shaker restrictions, manual moves.

Pipetting method and tip economy

Full columns use eight channels for 20 µL substrate, 20 µL toluene, 50 µL workup, 180 µL dilution and 20 µL sample transfers. Ligand and 10 µL Pd transfers use p20 single; partial columns use it too. The 20 µL reference ligand transfer is split into two 10 µL portions to leave room for the air gap and surplus. Eight-channel sample tips remain fresh for each column; every channel stays with its own well.

The initial recipe uses two source prewet cycles, slow aspiration, 0.5-second delays, a 2 µL air gap and 1 µL retained reverse-pipetting surplus. Fresh stock tips prewet in their stock source before visiting any destination. Fresh analytical sample tips use the separate same-well conditioning sequence below. Reused stock tips do not return the retained surplus to the source. Volume planning includes surplus losses. No blowout into the destination is used, because that would deliver an uncontrolled surplus.

`reuse_tips: true` means one tip can serve consecutive transfers of the **same stock** during the same stage, dispensing above the destination without touching it. No tips move between different ligands/stocks. Sample-transfer tips are never reused between samples. This requires a measured contamination and delivery check; disable reuse if contact, splash, droplet carryover or unreliable noncontact delivery occurs. The live exporter checks the validation setting. If reuse is disabled, additional tip-rack replacement prompts are expected; replace all racks for that pipette with fresh racks in their original slots, then resume.

DCM and toluene are volatile and air-displacement delivery may drip or lose volume. Validate the programmed speeds, prewetting and air-gap/reverse-pipetting method using the actual solvents and source/destination geometry. Check gravimetrically or with another suitable calibrated method, allowing for evaporation during weighing. Cap bulk reserves, minimize open-source time, and monitor the first transfers from each source. Visible bubbles or a liquid level too low for immersion require a pause and correction; repeating an aspiration does not restore a known dose.

Preserve the DCM stock concentration through first-round drying/assembly/heating: use validated individual source covers for unused reservoir wells, or HTE's equivalent handling procedure. The protocol pauses before dry-down and again before second-round predosing to check covers and source accessibility. Never leave a reservoir cover in a pipetting path; remove it before resuming aspiration. If concentration has changed through evaporation, a remaining liquid volume does not establish the stock concentration. Use a verified fresh aliquot from the preprepared reserve and record the refill; include sufficient reserve for that procedure in the stock-preparation calculation.

The code aspirates at `bottom(1 mm)` and dispenses at `top(+2 mm)`, so it never intentionally touches the bottom or recipient liquid. **These are starting heights for bench validation.** Confirm bottom clearance and adequate immersion at the lowest source volume, particularly in a shallow 100 µL glass extract. Check position offsets and vial seating. If a positive bottom clearance leaves the tip above the meniscus, change the validated height/source volume/geometry; do not set a negative clearance. Use a larger allowance if reservoir depletion exposes any of eight tips. The App's simulated liquid level is not proof of actual immersion.

During the clock: operator sequence

1. Start the timer with first-round dosing, after stocks, preselection, code installation and hardware/analytical validation are complete. Import only the live `round1/dose_and_react.py` with its matching CSVs into the local OT-2 App. Confirm offsets, sources and deck.
2. Dose 20 µL substrate/DCM into the 66 first-round wells in slot 3. Move the open block manually to heater-shaker slot 10. The initial dry-down program is 37 °C, 300 rpm and 10 minutes, **pending HTE verification**. Use the approved ventilation setup; elapsed time alone does not prove DCM removal. Inspect the dry endpoint, let it reach the handling condition and return it to slot 3.
3. Dose ligand stocks, toluene and finally Pd to reach 50 µL in each well. Seal using the validated block assembly. Move the sealed block onto the adapter and start 95 °C / 500 rpm. The programmed four-hour hold begins after the module setpoint is reached; use the validated liquid thermal-lag procedure.
4. While round one reacts, the OT-2 predoses the ten second-round substrate wells in the second array in slot 6. It writes no ligand choice into those wells yet. Remove that array at the pause to HTE's ventilated ambient-evaporation location. Resume promptly: this pause and predosing time are

subtracted from the programmed hold. **No second heater dry-down is scheduled.** Verify dryness before second-round catalysts are added. Keep well IDs/orientation intact, prevent contamination, and use the printed `round2_predose.csv`.

5 In parallel, prepare and run authentic-standard/calibration solutions on LC. Use the standards already purchased; never spike the authentic SM/product standards into reaction wells. Record retention references, area response and naphthalene response. Complete calibration before first-round samples reach the autosampler. Cap/remove dosing sources without altering required labware positions during active motion.

6 At the end of the hold, heating/shaking deactivate. Follow the validated thermal-handling SOP at the manual move prompt to transfer the sealed block off the heater to slot 3 and cool. Keep sealed until the validated opening temperature. If HTE's handling SOP requires white module status first, wait for it. Do not improvise hot-block handling to rescue the deadline.

7 Load the exact LC plate into slot 6 and start `round1/workup.py` with the cooled block in slot 3. Open under HTE's SOP. Add 50 µL workup solvent **without extra naphthalene**; manually move to the heater-shaker, shake 2 minutes without heating, and return to slot 3.

8 Confirm the validated homogeneous extract. Add 180 µL LC diluent to the mapped LC wells, then transfer 20 µL extract from the corresponding reaction wells. Seal and shake the LC plate for 2 minutes with HTE's validated plate/adapter method, then submit using `lc_sequence.csv`. The final LC-plate mixing is an operator step; do not assume the reaction block's adapter works with the LC plate.

9 Run all 66 injections. After the complete export is available, normalize it and publish its completion manifest as described below. The local watcher computes yields, checks QC, performs five fresh LLM scoring calls using all valid first-round data, and creates `next_design.csv` and `next_dosing.csv` for ten untested pairs.

10 Verify the ten predosed second-round wells are dry and match the manifest for 5 µmol substrate plus 0.050 µmol naphthalene. Refill source tubes/reservoirs from the preprepared reserves according to second-round `deck_loads.csv`; keep leftovers capped. Export/import the second-round live dosing protocol, which adds **zero new substrate or IS** and contains **no DCM heater step**. Assemble, seal and run its four-hour reaction, then perform the same cooling/workup/dilution/mixing procedure.

11 Run the final ten injections and quantify them. Stop the competition clock when those analyses are complete, as agreed. Save protocols, Opentrons JSON logs, all LC/calibration files, raw model responses and actual timestamps.

HTEL lists plate shakers among its equipment, but confirm which device and settings are available for the last LC-plate mix. If only the OT-2 heater-shaker is used, a validated LC adapter/plate swap must be added to the local operator procedure and its time counted. HTEL equipment.

Extraction and analytical calibration

The requested toluene + aqueous ACN workup may be biphasic. A fixed aliquot from one phase is not a known fraction of total product, and vigorous shaking does not guarantee a homogeneous sample. Before the campaign, HTE must demonstrate a homogeneous sampling matrix with known recovery for substrate/product/IS, or establish and document a quantitatively validated extraction alternative and update the configured volumes. Until then `homogeneous_sampling_validated` remains false. Visible solids or two phases stop sampling. Also verify analyte stability in the acidic workup and autosampler hold.

Representative aliquots and gentle liquid handling

Use this sequence for both rounds. Parameters below are proposed starting settings, not measured accuracy claims. Release them through the existing solvent-pipetting and homogeneous-sampling bench checks.

Action	Analytical handling rule
Cool and homogenize	Keep reactions sealed through cooling. After workup, use the validated two-minute mixing procedure and a consistent mixing-to-sampling interval. Remix under the approved method if delayed. A clear upper layer is not proof of quantitative recovery.
Condition each fresh sample tip	Before visiting the LC well, aspirate and return the transfer volume plus the planned surplus **twice into the same reaction well**, below the surface at the validated positive clearance. Each channel returns to its own well. No air gap or blowout during these cycles; allow liquid to settle. This conditions the tip; it does not validate an emulsion or dissolve solids.
Aspirate the aliquot	Hold the tip vertical, below the meniscus and above the bottom. Check immersion at the lowest remaining volume for every channel. Aspirate slowly, pause while immersed, then form the small air gap above liquid at a slow rate. A travel air gap is intentional; bubbles inside the measured liquid slug are a fault.
Dispense the aliquot	Add LC diluent first, then the measured extract, using the released noncontact position close to the well opening. Pause for droplet delivery. Discard the retained surplus with the sample tip; never blow it into the LC well. Never touch an LC well and return that tip to the reaction or solvent reservoir.
Mix and prepare for injection	Seal and shake the LC plate for two minutes. Let bubbles clear for a validated, consistent interval; inspect for precipitation. Filtration, centrifugation or taking only supernatant requires validated analyte/IS recovery. Confirm autosampler needle height and enough liquid for the injection, draw losses and any authorized repeats. Keep samples sealed at the validated hold temperature/time.
Recover an uncertain transfer	Quarantine that LC well. Use a new tip and fresh LC destination with the complete correct diluent load and a new aliquot of the rehomogenized extract, provided sufficient extract remains. Preserve the failed preparation and its record. Never guess a top-up volume, add more IS, or restart the full workup to repair one well.

The workup protocol now separates sample settings from stock dosing: `sample_prewet_cycles = 2`, `sample_aspirate_height_mm = 1`, `sample_flow_rate_multiplier = 0.5`, `sample_air_gap_rate = 0.1`, and `sample_pause_seconds = 1`. Thus sample liquid flow starts at 2/4 µL/s aspirate/dispense on p20 and 7.5/15 µL/s on p300; air-gap aspiration uses one tenth of that aspiration rate. Prewet returns have no net added volume. Account for the surplus discarded by each fresh tip, including split p20 transfers. Keep these fields in the campaign-specific configuration; regenerate exports and revalidate handling after changes. Update measured workup timing for the extra cycles/delays.

Before the clock, challenge the complete extraction/dilution with known low/mid/high analyte amounts in the actual matrix. Compare independently prepared reference dilutions and repeated preparations, including all eight channels and near-depleted source conditions; record bias, variability and analyte/IS recovery against HTE's method criteria. The 20 µL transfer is small relative to the p300 capacity: validate it specifically. If inadequate, revise the aliquot/dilution or use a reviewed p20 sampling plan and account for its time/tips. IS ratios can compensate for common injection/dilution variation but cannot prove selective analyte recovery, IS survival or absence of coelution.

Prewetting, delays and air-gap settings need solvent-specific validation; reverse pipetting is not inherently more accurate for volatile liquids. Opentrons volatile-liquid study, API liquid-handling commands. Thermo identifies low sample volume, incorrect needle height and trapped air as causes of poor area precision. Vanquish troubleshooting.

The proposed 20 µL + 180 µL dilution is tenfold from a nominal 100 µL homogeneous extract. At 100% yield, final product is 5 mM; final naphthalene is 50 µM. This helps eight-channel throughput but may be too concentrated for the chosen LC/MS response or injected toluene content. HTE must validate separation, injection solvent, calibration range and absence of detector saturation/ion suppression. It is not a finalized LC dilution solely because it fits a plate.

If 5 µL + 195 µL is required, set those volumes and regenerate all files. It is a fortyfold dilution and uses p20 single for samples. The preloaded 0.050 µmol naphthalene now gives 12.5 µM final IS; update `internal_standard_final_uM` and prepare standards at that concentration. The code rejects inconsistent IS concentrations. Change aliquot/dilution only before the complete campaign is frozen, or explicitly record a new validated configuration with calibration matching the actual samples.

Prepare authentic substrate/product calibration stocks in an LC-compatible solvent at measured concentrations; weigh with COA corrections. A proposed 100 mM stock is 16.116 mg/mL PyFluor or approximately 9.709 mg/mL 2-fluoropyridine, subject to solubility/stability verification. Stocks can be prepared before the clock; working standards and injections can be prepared during the first hold. Naphthalene needs a detector where its response is reliable.

Use five nonzero fitted levels of each analyte: 100, 500, 1,000, 2,500 and 6,000 µM, each with **fixed 50 µM naphthalene**, plus an IS-containing zero-analyte blank. Add an independently prepared 5,000 µM check excluded from the fit. Match the validated final sample solvent/matrix and IS concentration, including the solvent introduced by the analytical stocks. Use the lower working-stock concentration for the lowest levels. The preparation table avoids sub-microliter transfers. Standards above the validated linear range require a revised dilution, not extrapolation.

The default 14-injection sequence contains five nonzero levels, a zero-analyte blank and an independent check for each of SM and product. The single-analyte series and IS-containing blanks also provide retention identity references, subject to LC/MS confirmation. All 14 are scheduled during the first four-hour hold. In `calibration_template.csv`, rows marked `calibration` enter the fit and rows marked `check` are assessed independently, with a provisional ±15% relative concentration tolerance. Any additional identity, blank or repeat injections must be added to the timing configuration.

Adding IS before the reaction requires HTE to verify naphthalene retention through both dry-down methods, the sealed hot reaction and extraction, and to check chemical interference. Loss of IS can inflate area-ratio yields. The software assumes quantitative recovery and does not apply an unmeasured correction. Live export requires `pre_reaction_internal_standard_validated`, as detailed in the stock instructions.

For each analyte fit `area_analyte / area_IS = slope × final_concentration_µM + intercept`. Substrate and product have separate response factors. Do not use area%, TIC% or an uncalibrated MS area as yield. The current campaign requires five distinct nonzero concentrations, a measured zero-analyte blank and at least one independent check per analyte, positive slopes and the configured fit threshold. R^2 >=0.995 is a software gate, not complete method validation; verify accuracy, precision, blanks, LOQ, recovery and linearity with HTE. Set measured LOQs and nonoverlapping retention windows from authentic standards. MS confirmation must support the assigned analyte peak, especially if a short method coelutes other components.

If HTE's complete integrated peak list omits nondetected analytes, set `missing_analyte_policy` to `censor` **only after verifying that export behavior**. A valid IS and completed injection/export are required. The reader then reports `yield_upper_bound_percent`, not zero; below-LOQ detected peaks are also represented as bounds. The default is `reject` because a missing row could mean a bad export. Multiple peaks in the reference window, absent/invalid IS, inconsistent method, out-of-range calibration or implausible mass balance are QC failures. Invalid numerical results are removed before LLM feedback. With the current zero-tolerance batch QC setting, resolve a failed measurement/export before the loop proceeds; reaction failure itself is valid data when analytically measured.

Repeat injections and repeat preparations

A second injection from the same LC well checks injection/integration repeatability; a new independent dilution from the reaction extract checks sample preparation. Neither is a repeat reaction. During pre-campaign validation, check needle-wash effectiveness and inject a matrix blank after a high standard to measure carryover. Use HTE-approved criteria for area-ratio agreement, recovery and blank response. Configure the actual wash/needle settings; do not copy settings from another autosampler. Thermo carryover guidance.

During the competition, remeasure only when QC or a documented instrument/transfer fault requires it. Record the original and replacement IDs, reason, injection order, dilution and acceptance decision. Retain all raw results; choose the valid record by the predefined QC rule, never by highest yield. The quantitative export must contain one accepted result per original reaction sample ID; the current reader rejects duplicate/ambiguous peaks and does not average replicates automatically. Review discrepancies before publishing a completion marker. Every added injection costs its measured full cycle, at least two minutes here, and belongs in the timing plan; these checks are not silently added to the 14-standard or 76-sample sequence.

Export contract and automatic second-round decision

Set `analytics.method_id`, `channel`, `references`, `quantification_limit_uM`, `column_mapping` and, for XLSX, `export_sheet` to actual HTE values. `references` must have `substrate`, `product` and `internal_standard`, each with `retention_time_min` and `window_min`. Do not fill these from the demo. The software/instrument public listings are descriptive provenance, not active method identifiers.

Example mapping structure, with placeholders replaced by exact Chromeleon export headers:

```
{
  "sample_id": "EXACT_SAMPLE_HEADER",
  "retention_time_min": "EXACT_RT_HEADER",
  "area": "EXACT_AREA_HEADER",
  "channel": "EXACT_CHANNEL_HEADER",
  "method_id": "EXACT_METHOD_HEADER"
}
```

The normalized CSV has one row per peak:
`sample_id,retention_time_min,area,channel,method_id,qc_flag`. Retention times are minutes. `sample_id` must exactly match `R1-001` etc. from `lc_sequence.csv`; retain full peak lists on the selected quantitative channel. Other channels are ignored, never summed. Per-channel spreadsheets that omit channel/method columns need those columns added from verified acquisition metadata before normalization. The selected channel must not change between calibration and samples.

`calibration.csv` follows `hte_inputs/calibration_template.csv`: sample ID, role (`calibration` or `check`), authentic analyte label (`substrate` or `product`), final concentration, its integrated area, IS area, final IS concentration, method and channel. The two analytes may come from the same validated mixed-standard injection; include a row for each and update the injection count. Use the supplied template with explicit roles; a legacy CSV without independent `check` rows fails the calibration gate.

Prepare the watcher before first-round LC finishes:

```
python -m hte.cli --config local_data/campaign.json watch --inbox output/hte-lc-inbox \
  --dosing output/hte-round1/dosing.csv --calibration output/hte-calibration.csv \
  --output output/hte-iteration
```

After all 66 injections have finished, run from a second terminal (or a verified Chromeleon post-sequence hook):

```
python -m hte.cli --config local_data/campaign.json normalize path/to/complete_vendor_export.xlsx \
  --output output/hte-lc-inbox/round1.csv
```

```
python -m hte.cli --config local_data/campaign.json \
  complete-export --peaks output/hte-lc-inbox/round1.csv \
  --dosing output/hte-round1/dosing.csv \
  --output output/hte-lc-inbox/round1.ready.json
```

The `complete-export` command is an operator attestation that every expected injection has finished. A stable CSV alone does not establish sequence completion. If the export is already normalized, place it directly at `round1.csv`, then publish the manifest. The manifest verifies expected sample IDs, method and SHA256 file hashes. The watcher responds only to `*.ready.json` with a matching CSV; partial, mismatched or changed exports cannot trigger a valid batch. Input hashes are checked before and after ingestion, and the completed iteration hashes its analysis, scoring and handoff files. Publishing the same event twice does not repeat paid calls; changed inputs or archived outputs are rejected. Failed events are recorded once; correct the data/configuration and publish a corrected event instead of repeatedly charging requests.

Restart the watcher after changing its configuration, inventory or evidence, because those inputs are loaded when it starts. Calibration/export files are read for the event. Set the API key before starting the watcher.

The output directory is printed as `output/hte-iteration/iteration-....` Inspect `analysis/results.csv`, `analysis/qc_report.json`, `scoring/ranking.csv`, `next_design.csv` and `next_dosing.csv`. The second round uses only LLM mean-score ranking of the 400 untested pairs, incorporating first-round calibrated observations and the single reference. It does not use an exploratory split or GoLLuM acquisition.

Live protocols are generated automatically when the bench validations, analytical settings and measured timing pass the release checks; otherwise the next dosing CSV is still produced for review. Explicit export uses:

```
python -m hte.cli --config local_data/campaign.json \
  export --design output/hte-iteration/iteration-REPLACE/next_design.csv \
  --history output/hte-round1/dosing.csv \
  --output output/hte-round2 --live
```

For round two, `--history` must point to the actual first-round dosing file; repeated pairs or edited dose amounts stop export. Use a new output directory each time, including after a partial export, to preserve earlier files. Validate the local App analysis and compare sample IDs, wells, substrate predose and stock loads before running. The watcher does not start the robot or upload protocols remotely.

For final second-round results, normalize/publish a manifest in the same way, then:

```
python -m hte.cli --config local_data/campaign.json analyze --dosing output/hte-round2/dosing.csv \
  --peaks output/hte-round2-peaks.csv --ready output/hte-round2-peaks.ready.json \
  --calibration output/hte-calibration.csv --output output/hte-final-analysis
```

Optional MOCCA integration

HTE's standard integrated peak exports are the default analytics path. MOCCA2 is available when **raw HPLC-DAD time × wavelength data** can be exported, with authentic standards processed through the same pipeline. It does not reconstruct chromatograms from peak-area CSVs and does not process ISQ MS spectra. Native Chromeleon raw data are not assumed to match MOCCA's parsers. Provide a supported raw export or a verified conversion to the time/wavelength matrix format used by the tested adapter. MOCCA2 documentation.

`mocca_manifest_template.json` maps raw paths to sample IDs. For a ChemStation-style matrix CSV, the first row is wavelengths in nm, first column is time in minutes, remaining cells are detector responses, and encoding is UTF-16. Paths are relative to the manifest. The included settings are a starting template, not a validated method:

```
python -m hte.cli mocca --manifest path/to/raw_manifest.json \
```

```
--settings hte_inputs/mocca_settings.json --output output/hte-mocca
```

This calls actual MOCCA baseline correction, peak detection and Fraser-Suzuki deconvolution, reports physical time integrals at the selected wavelength, and marks unresolved peaks as QC failures. Test cases use synthetic raw DAD data. Calibration areas must be generated with exactly the same preprocessing/units, not copied from Chromeleon if their integration definitions differ. Confirm the required detector channel matches the adapter's emitted name, e.g. `MOCCA_DAD_254nm`.

Timing and validation record

At two minutes **injection-to-injection**, 66 + 10 reaction samples require 152 minutes of LC time. The reaction holds require 480 minutes. That leaves 88 minutes within 720 for every other step. Fourteen standard injections use 28 minutes inside the first hold, so they add no serial time if completed then. Predosing and ambient second-round DCM evaporation also occur inside that hold.

The supplied provisional overheads are 30 minutes dosing, 10 first-round evaporation, 12 heating, 12 cooling, 20 workup, 8 decision and 5 other: 97 minutes, giving **729 minutes / 12 h 9 min**. This is an honest planning warning, not a verified duration. Heavy-block heating/cooling, two-minute LC overhead, noncontact delivery and API quotas can change it materially. Replace estimates with rehearsal measurements; achieve at least nine minutes of documented savings, plus a contingency margin. Prioritize off-module cooling under HTE's approved handling SOP, full-column workup/sampling, validated source-dedicated tip reuse, preconfigured LC sequences and pretested concurrent scoring. Do not shorten the specified four-hour reaction or skip method validation to make a nominal timeline fit.

`python -m hte.cli check` prints the timing and outstanding bench fields. Before live export, HTE records successful checks in `validation`: chemistry, block/adapter/seal, solvent pipetting, heating/evaporation, homogeneous sampling, LC calibration, local App simulation, ambient second-round evaporation, pre-reaction IS retention/compatibility, and noncontact tip reuse when enabled. Live export also requires actual LC identifiers, nonoverlapping retention windows, measured LOQs and a configured schedule within the deadline; the template at 729 minutes cannot be released. These fields represent completed bench work; changing a Boolean is not validation. A changed plate, solvent, dilution, stock concentration or height requires review of the affected check and regenerated exports.

Keep a campaign folder containing inventory/COAs, configuration, embedding provenance, all scoring responses, designs, dosing/deck/stock CSVs, exact labware JSONs, App analysis and Opentrons JSON logs, standards/LC exports/raw data, QC reports and actual timestamps. The offline demo and tests establish software behavior; they cannot establish chemical success or a 12-hour physical run.

3. Combined stock and analytical calibration

Editable source: docs/stocks-and-calibration.md

The reaction stock contains starting material and naphthalene together. A separate reservoir channel holds the LC dilution solvent. Calibration solutions use separate authentic analyte stocks and a pure analytical naphthalene stock, so analyte concentration can vary while the reference concentration stays fixed.

Reaction stock, container and deck source

Prepare **5.00 mL of 250 mM PyFluor + 2.50 mM naphthalene in DCM** before the clock. At the nominal molecular weights and 100% assay, this contains **201.45 mg PyFluor and 1.6021 mg naphthalene**. The stock planner reports both solutes, with separate assay corrections under `substrate_DCM` and `naphthalene`. Bring the solution to its final volume; adding 5 mL solvent to solids is not a final-volume preparation.

Avoid weighing 1.6 mg directly unless the balance supports that measurement. One preparation route is:

1. Prepare a separate **50.0 mM naphthalene master in DCM**, for example 12.817 mg in a final 2.00 mL, correcting for the applicable assay. Use a larger scale if the balance requires it.
2. Weigh the corrected PyFluor mass into a **20 mL glass vial with a DCM-compatible PTFE-faced closure**.
3. Add **250 µL of the 50.0 mM DCM master**, dissolve in DCM, and bring the combined solution to **5.00 mL final volume** using validated volumetric handling.
4. Label both concentrations, solvent, assays, lots, preparation time and expiry/recheck conditions. Store the reserve capped off deck.

The vial is the preparation/storage container. The pipetting source is **A1 of the 12-channel reservoir in OT-2 slot 2**, with nominal 15 mL capacity per channel. It is not a Falcon tube. All eight tips can aspirate from the same long channel to dose full columns. The default source-dedicated tip-reuse plan loads approximately **2.54 mL** for the 66 first-round wells and ten second-round predoses, including the provisional 1 mL aspiration allowance. Fresh-tip mode requires approximately 2.60 mL. **Use the generated** `deck_loads.csv` **for the exact campaign load**, and retain the rest of the prepared 5 mL capped.

The library reservoir is a geometry placeholder. HTE must confirm the actual reservoir and DCM contact compatibility for the short dosing interval. The NEST product specification identifies a polypropylene 12-channel reservoir; a matching footprint alone does not establish solvent compatibility. Keep the bulk DCM reserve in glass and do not assume a polypropylene centrifuge tube or reservoir is suitable for prolonged storage.

Each **20 µL** combined-stock dose supplies **5.00 µmol substrate and 0.050 µmol naphthalene**, equivalent to 1 mol% naphthalene relative to substrate. Evaporate DCM using the validated first-round procedure. The ten second-round wells receive this same combined stock during the first hold and undergo ambient evaporation; their second-round catalyst protocol adds neither substrate nor naphthalene again. Both amounts appear in `dosing.csv` and `round2_predose.csv`.

Workup and sample dilution

Reservoir **A4** holds the workup solvent: 1:1 water/ACN with 1% v/v HCOOH, **without additional naphthalene**. Add 50 µL per reaction and mix for two minutes after cooling/opening under HTE's SOP.

Reservoir **A5** holds the LC diluent: the validated 1:1 water/ACN with 1% v/v HCOOH, **without naphthalene or analyte**. Dispense 180 µL into each LC destination first, then transfer 20 µL of

homogeneous extract with fresh sample tips, and mix the LC plate for two minutes. Full columns use eight channels for both solvent dispensing and sample transfer. Fresh sample tips condition only in their own reaction well before transfer, using the sample-specific slow-flow settings. Follow the runbook sampling sequence for immersion, bubble checks, surplus disposal and failed-transfer recovery; repeated aspiration cannot make a two-phase extract quantitative.

With quantitative retention/recovery, 0.050 µmol IS in a nominal 100 µL extract is 500 µM. The 20 + 180 µL dilution gives **50 µM naphthalene** and, at 100% yield, **5,000 µM product**. Sample dilution reduces both analyte and IS concentration; their concentration ratio stays constant. Changing the aliquot to 5 µL plus 195 µL gives 12.5 µM IS and a 1,250 µM maximum product concentration. It also changes sample transfer to p20 single. Update the configuration, standards and generated files together; the code checks their IS concentration agreement.

Separate analytical stocks

Prepare these before the clock, then prepare/run the working standards while the first reaction runs. Use the actual bottle/COA molecular weights and an HTE-approved LC-compatible solvent. ACN is a proposed stock solvent, subject to solubility and method validation.

Analytical stock	Proposed final volume	Nominal preparation at 100% assay
Authentic PyFluor, 100 mM, no IS	2.00 mL	32.232 mg, MW 161.16
Authentic 2-fluoropyridine, 100 mM, no IS	2.00 mL	Approximately 19.418 mg, MW 97.09; verify bottle
Pure naphthalene, 50 mM, in LC-compatible solvent	2.00 mL	12.817 mg, MW 128.17
Each analyte working stock, 10 mM, no IS	1.00 mL each	100 µL of its 100 mM stock, bring to 1.00 mL
Pure analytical IS working stock, 1 mM	2.00 mL	40 µL of the 50 mM analytical master, bring to 2.00 mL

Keep the **DCM naphthalene master for reaction dosing** and the **LC-compatible naphthalene master for calibration** distinctly labeled. Do not introduce DCM into the calibration series through an unvalidated solvent substitution. Do not add authentic SM/product calibration solutions to reaction wells.

Five nonzero analyte levels with fixed reference

Prepare one series for SM and one for product. Each standard below has **1.00 mL final volume**, **50 µM naphthalene**, and the indicated known analyte concentration. The five nonzero fitted levels are 100, 500, 1,000, 2,500 and 6,000 µM. Also prepare an IS-containing zero-analyte blank and an independently prepared 5,000 µM check for each series. This gives **14 injections**, scheduled during the first four-hour reaction. These analytical blanks/checks do not occupy additional reaction wells.

Final analyte, µM	Analyte stock	Analyte volume, µL	1 mM IS, µL	Remaining solvent volume to final 1 mL, µL	Role
0	None	0	50	950	Zero-analyte calibration blank
100	10 mM	10	50	940	Calibration level 1
500	10 mM	50	50	900	Calibration level 2
1,000	10 mM	100	50	850	Calibration level 3
2,500	100 mM	25	50	925	Calibration level 4
6,000	100 mM	60	50	890	Calibration level 5
5,000	100 mM	50	50	900	Independent check, excluded from fit

The remaining solvent must be adjusted for the solvent introduced by both stocks to obtain the **same validated final matrix across standards and samples**. This table specifies solute amounts and volume arithmetic; HTE must supply the final matrix recipe after assessing phase behavior, recovery and the LC method. Adding the same nominal 1:1 diluent to every row without accounting for variable stock solvent would change the matrix. These standards are unsuitable for live quantification until that method validation is complete.

Use the sample IDs and roles in `hte_inputs/calibration_template.csv`, replacing empty area/method/channel fields with measured values. A blank analyte area of zero must represent a verified blank measurement, not a missing export. The fitter uses the blank and five nonzero levels; rows marked `check` remain outside the fit. Prepare levels independently from the measured master/working stocks, fully mix and cap each tube, and use fresh tips between standards. Prepare the held-out check as a separate dilution, with an independent stock preparation when feasible; it must not be an aliquot of a fitted standard. Keep dilution labels and volume records. The proposed independent-check acceptance is ±15% relative concentration error, configurable under `analytics.calibration_check_tolerance_fraction`; HTE must approve this criterion and the fitted range/LOQs for the actual detector.

Serial dilution of the combined reaction stock alone cannot make these fixed-IS standards: it keeps the SM:IS ratio at 100:1 and lowers both concentrations together. At 50 µM naphthalene it would supply 5,000 µM SM. Separate authentic analyte and pure analytical IS stocks let HTE vary the analyte while holding naphthalene at 50 µM.

Validation of IS added before the reaction

HTE must verify naphthalene retention/recovery through first-round heated DCM removal, second-round ambient DCM removal, the sealed 95 °C hold and the complete extraction/dilution. Confirm that this low loading does not alter the intended chemistry or interfere with the selected peaks. Naphthalene has a reported nonzero vapor pressure; its survival through an open dry-down is an experimental question. Manufacturer properties.

IS loss lowers its peak area and can overestimate analyte yield from the area ratio. The software has no measured loss correction; a detector-area check alone cannot establish quantitative recovery. Record the completed bench assessment in `validation.pre_reaction_internal_standard_validated`. Live export remains gated until it and the other applicable validations pass. Any revised IS identity, loading or addition stage requires consistent chemistry context, calibration, stock plan and regenerated protocols.

4. Facts to confirm with HTE

Editable source: docs/hte-questions.md

Record the confirmed values in a campaign-specific copy of `hte_inputs/campaign.json` before generating live protocols. This checklist accompanies the operator runbook.

Analytical method and export

- Exact two-minute method ID and measured injection-to-injection cycle, including overhead.
- Column, gradient, injection volume, quantitative DAD wavelength/channel and retention windows for substrate, product and naphthalene.
- Representative CSV/XLSX export: headers, sample/well mapping, retention-time and area units, method/channel fields, nondetections and failed injections.
- Complete-sequence export or an automatic completion marker for the local watcher.
- Raw DAD export format if optional MOCCA processing will be used.

Public HTEL equipment listings identify Vanquish Horizon Duo, ISQ-EM and DAD, and UZH lists `LC-ISQ-HTL-01` in Chromeleon. These listings do not establish the actual method or quantitative channel. Equipment, Chromeleon.

Calibration and sample compatibility

- Schedule authentic standards, calibration series, blanks and checks during the first reaction hold.
- Validate extraction/recovery, analyte stability and phase homogeneity after adding 50 µL of 1:1 water/ACN with 1% formic acid, without extra IS, to 50 µL reaction mixture.
- Verify retention/recovery and chemical compatibility of 0.050 µmol naphthalene added with SM before DCM evaporation, including both dry-down methods and the 95 °C hold. Confirm the five nonzero analyte calibration levels, fixed 50 µM IS and independent-check acceptance; see the stock and calibration instructions.
- Confirm the proposed 20 µL aliquot into 180 µL diluent is appropriate: nominal maximum product concentration is 5 mM and naphthalene is 50 µM.
- Confirm final matrix, dilution, injection concentration, plate/seal, minimum volume and any filtration or centrifugation.

Glass reaction block and heating

- Exact glass block, insert and seal, with a measured Opentrons labware definition.
- Secure mounting on the OT-2 heater-shaker universal flat adapter, latch/bolt clearance, allowable shaking mass and open/sealed travel clearance.
- Toluene loss, sealing and SO2 compatibility at 95 °C for four hours.
- Actual liquid thermal lag, off-module cooling procedure and safe thermal handling.
- A second vial array or compatible removable tray for predosing ten substrate wells during round one.

The Para-Dox Gen II 104960 is a candidate with 1 mL inserts and a standard microplate footprint. Its footprint does not establish module fit or thermal performance. Drawing.

Stocks and solvent delivery

- Bottle/COA identities and assay values for all ligands, Pd(COD)(DQ) and substrate.
- Homogeneous combined 250 mM substrate + 2.5 mM naphthalene in DCM, 30 mM ligand/toluene and 60 mM Pd/toluene stocks at dispensing temperature.

- Confirm the capped 20 mL glass preparation vial for the 5 mL SM/IS stock and short DCM contact compatibility of the on-deck 15 mL reservoir channel. Keep a separate channel for LC dilution solvent.
- Glovebox preparation/storage for ligands, labeled stocks and sufficient capped reserves.
- Source containers, dead volumes, aspiration/dispense heights and validated organic-solvent delivery settings.
- p20 single GEN2 and p300 eight-channel GEN2 configuration; source-dedicated noncontact tip reuse only after validation, with fresh sample tips.
- First-round DCM evaporation on the heater-shaker; second-round ambient evaporation under the approved ventilation setup, with a dry endpoint check before catalysts.

Timing and local operation

- Rehearse dosing, heating, thermal equilibration, cooling, workup, manual moves, LC handling and the second-round LLM decision.
- Confirm uninterrupted LC and local OT-2 App availability and access to a plate shaker for the analytical plate.
- Confirm permitted Anthropic access from the operator workstation and test model access before the competition.
- Stocks and initial preselection are outside the clock. The clock stops after the ten second-round analyses.

Two four-hour holds and 76 analyses at two minutes each consume 632 minutes, leaving 88 minutes for all other operations. The provisional complete schedule is 729 minutes and therefore requires improvement supported by rehearsal measurements.

5. Claude prompt caching and costs

Editable source: docs/prompt-caching.md

Prompt caching is enabled by default for both scoring campaigns. The API key and first-batch commands remain the same. Configuration is in `hte_inputs/campaign.json`, under `llm.prompt_cache`:

```
{
  "enabled": true,
  "ttl": "1h",
  "read_multiplier": 0.1,
  "write_5m_multiplier": 1.25,
  "write_1h_multiplier": 2.0
}
```

The one-hour lifetime allows repeats and queued requests to reuse the shared input. The second campaign begins after the first reaction and LC sequence, so it normally writes a new cache entry. This does not preserve the initial cache across the four-hour reaction.

Request layout

The fixed system instruction is followed by three user content blocks:

1. Fixed reaction conditions, objective, all ligand IDs/names/SMILES, the explicit pair catalog, and CAS-matched single-ligand SI yields with a brief condition label (unreported values are null), with an explicit `cache_control` breakpoint.
2. The complete measured feedback shared by the campaign's repeats, with another explicit breakpoint.
3. The shuffled candidate IDs and exact candidate count, with no cache breakpoint.

The current prompt requests scores only and explicitly disables extended thinking. Changing the prompt, schema or thinking configuration creates a different cache prefix, so old v3 cache entries are not reused. The fixed blocks are serialized identically across repeats. Output-schema property ordering is canonical and remains identical for the same candidate cohort: schema changes can invalidate Anthropic's prompt cache. With smaller configured batches, cohorts stay fixed and their internal candidate order is shuffled independently for each repeat. The default still scores all 465 first-round candidates, or all 400 remaining second-round candidates, in each request.

The first real scoring response for each schema completes before its followers are submitted. It warms the cache without an additional paid warm-up request. Subsequent scoring requests can run in parallel. All five responses remain separately generated; previous scores or assistant answers are not supplied as cached context.

Cost and verification

For the configured Opus 4.8 prices, one-hour writes cost 2× normal input, five-minute writes cost 1.25× and reads cost 0.1×. These multipliers are explicit configuration values. The preflight assumes every pending request misses cache, charges input at the selected write rate and reserves one retry. It does not rely on hoped-for cache savings to stay within the estimated $20 cap.

Each response record saves the provider's usage counters and a separate cost breakdown for uncached input, cache writes, cache reads and output. Detailed five-minute/one-hour write counters are used when available; otherwise the configured TTL is recorded as the pricing assumption. `input_tokens` excludes cached reads and writes, so those amounts are added separately.

After each campaign, inspect `scoring/cache_summary.json`:

- `cache_read_input_tokens > 0` and `cache_hit_requests > 0` verify reported reuse.
- `cache_creation_input_tokens > 0` with no reads means entries were written but reuse was not reported.
- Both counters zero means no cache activity was reported. Check prompt size, TTL and request consistency before a later campaign; the software does not buy extra retries solely to obtain cache hits.

The score command prints the cache status and hit counts. Cache summary costs estimate the returned responses using the configured rates; SDK retries whose usage is not returned cannot be itemized. The budget preflight reserves for them.

The mocked tests and offline rehearsal verify request layout, stable schemas, scheduling and accounting. Separate live four-pair smoke tests verified provider-reported reuse; the current v4 second request read 5,174 cached tokens, as recorded in the rehearsal evidence. **Live provider cache hits require a real API run** and are not established by the rehearsal. Anthropic documents a 1,024-token minimum for Opus 4.8; the normal full-inventory context is intended to exceed that threshold.

Primary references: Anthropic prompt caching, structured-output cache invalidation.

6. Software rehearsal evidence

Editable source: docs/rehearsal-results.md

The offline first-feedback-loop rehearsal completed successfully. All chemistry results and Claude responses in this exercise were synthetic; no paid API call or physical robot run occurred.

Check	Observed result
Real frozen T5 embeddings loaded	465 unordered pairs, 768 dimensions
Initial scoring request contract	Five mocked requests, each covering 465 pairs
Initial batch	65 distinct pairs plus one tBuBrettPhos reference
Mandatory pair coverage	All 31 ligands represented; minimum one pair well per ligand, with the single reference excluded
LC processing	66 calibrated synthetic results; batch QC passed
Feedback scoring request contract	Five mocked requests, each receiving all 66 results and scoring 400 untested pairs
Next dosing CSV	Ten new pair assignments using only LLM scores, with no pair repeats
Predosed substrate	All ten second-round substrate doses remain zero
Predosed internal standard	0.050 µmol naphthalene per well retained in the plan; no second substrate/IS stock or workup IS dose
Calibration	Five nonzero levels plus an IS-containing blank per analyte; independent checks excluded from each fit
Incomplete LC export	Ignored until the completion marker was published
Repeated completion event	Ignored; no additional scoring calls
Physical protocol generation by watcher	Disabled because bench validations remain incomplete
Optional GoLLuM execution	Actual attributed upstream optimizer, real embeddings, 61 uncensored synthetic pair priors, 465 finite acquisition values
Local API key file	Mocked scoring reads `anthropic.key`; environment takes precedence; malformed entries stop without echoing contents; no key saved in outputs
Local acceptance suite	101 passed, zero failures/errors/skips

Adversarial selector tests also cover a ligand whose every pair is below the Claude percentile floor, tight coverage within 16 wells, impossible coverage, and a manually edited design where the L17 reference is present but L17 is absent from all pairs. Coverage overrides are audited; missing pair coverage blocks export. The current offline loop uses the same mandatory coverage rule and completes all 31 ligands without extra wells.

The repository audit added regression checks for calibration roles/levels/blanks, CSV/XLSX normalization, stale dosing, repeated pairs, ties, live LC/timing gates, file races, concurrent processes and interrupted API calls. No additional paid requests were made. See the audit record.

The campaign rehearsal also rejects any call to a GoLLuM yield optimizer and verifies that every second-round assignment uses the LLM selection method. GoLLuM-inspired initialization needs no pair yields.

The historical optional GoLLuM check did not change the second-round design. Its local GP adapter trained a projection over frozen embeddings, keeping the language-model weights unchanged.

To reproduce the feedback loop from the repository root:

```
python -m hte.cli rehearse-loop --output output/fake-first-loop
```

Use a fresh output directory. `rehearsal_report.json` records the checks and the relative location of `next_dosing.csv`. The generated rankings use an explicitly labeled arbitrary emulator; they are unsuitable for deciding real experiments. The real initial selection requires the operator's Claude API key and the first-batch setup.

Separate live API checks

On 9 October 2026, three small Opus 4.8 smoke tests each scored four pairs twice. The current scores-only v4 prompt returned valid complete JSON with no reasoning or hypothesis field; both provider responses contained only text blocks with extended thinking disabled. The second response reported **5,174 cached read tokens**. The v4 test cost approximately **$0.0570**, bringing total estimated returned-usage spending across six generation requests to **$0.4023**, below the $1 smoke-test allowance.

The earlier v2/v3 tests used a different prompt and output contract; their $0.3453 combined cost is included in that total. A conservative preflight stopped one earlier v3 attempt before generation. The scores-only offline feedback rehearsal was rerun successfully: all 66 synthetic first-round results reached the prompt, 400 untested pairs were scored in five mocked calls and ten new pairs were exported without substrate/IS redosing.

These checks establish live model access, structured-response parsing and cache reuse for the smoke requests. The full 465-pair initial campaign has not been generated, and the model scores are not measured chemistry yields. The prompt guide supplies the identities, SI evidence and output contract. Raw responses and keys remain outside the public handoff.

Repository audit

The 9 October review checked all source modules, tests, campaign inputs, documentation and generated handoffs. The 31 ligand identities were compared with the original workbook; molecular weights and phosphorus counts were checked from the structures. Re-extraction of the 18 same-substrate single-ligand SI yields reproduced the stored evidence. The exact first-round prompt preview matched the published prompt; its scores-only contract and prompt caching remain unchanged.

The audit corrected these operational issues:

- Live export now requires complete analytical settings and a measured schedule within the competition deadline, alongside bench validations.
- Designs and imported dosing files are checked for the configured reference, unique pair identities, ligand coverage and all calculated quantities. Live round-two export requires the actual first-round history and rejects repeated pairs.
- Calibration requires the configured five nonzero levels, measured zero blanks and independent checks. CSV/XLSX import rejects incomplete exports while preserving measured zero peak areas.
- Equal scores/distances receive equal average ranks. Initial selection remains 50/50 Claude and embedding diversity; feedback selection remains entirely Claude.

- Process locks prevent duplicate local jobs. Attempt records prevent automatic repayment after a charged request is interrupted. Completed input/output hashes detect changes, and existing or partially written protocol exports cannot be overwritten.
- Production commands consistently use the campaign-specific configuration. The staff checklist remains two pages; the complete handbook retains the technical reference.

At the repository audit, all **88 tests passed**, including actual Opentrons simulation and the offline LC-to-Claude feedback loop. Additional direct demo and feedback rehearsals use synthetic results and mocked API responses. No new paid calls were made for this audit. The historical live smoke-test total remains approximately $0.4023.

The full real first-batch ranking has not been generated. No physical chemistry or hardware run was performed. HTE must still validate glass hardware, solvent handling, IS recovery, extraction, the LC method and actual timing. The template's **729-minute** schedule exceeds the **720-minute** limit and is deliberately blocked from live release until measured timings fit.

Analytical handling update

The subsequent handling update passed **101 tests**. Added OT-2 simulations verify that conditioning cycles return only to the original sample well, the net sample withdrawal includes just the intended aliquot and discarded surplus, LC delivery matches the planned aliquot, sample tips remain fresh and the liquid/air-gap flows use the sample settings. Full-column p300 and partial-column p20 transfers are checked. Disabling sample conditioning and invalid settings are tested; a transfer that exhausts the nominal extract is rejected.

The runbook now distinguishes reinjecting the same preparation from making an independent dilution, specifies bubble/phase/clearance checks and requires documented recovery from uncertain transfers. No paid API call or physical chemistry run occurred. The added conditioning and pauses need measured workup timings and matrix-specific bench validation before release.